

Recycling of Lignin-Based Biocomposites: Improving Sustainability and Enhancing Material Strength

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Abstract

Biopolymer-bound soil composites (BSC) are a novel class of cement-free building materials using biopolymer binders, many of which are sourced from the waste streams of major industries. This study investigates the recyclability of one particular BSC that uses kraft lignin as the biopolymer. Re-manufacturing of BSC was accomplished by mechanical disruption of the virgin material, followed by re-introduction of solvent, remixing, and remolding. We found that the compressive strength of recycled lignin-based BSC was higher than BSC made with virgin ingredients. To understand the microstructure of lignin-based BSC, a series of X-ray micro-CT images of the test articles were obtained. Images produced by the micro-CT method reveal differences in the microstructure of the re-manufactured specimens indicating an enhancement of the association between lignin and aggregate particles. This study demonstrates the feasibility of recycling BSC and provides insight into the importance of biopolymer-aggregate association in determining the mechanical properties of BSC.

Keywords: Lignin, Biopolymer-bound soil composites, Sustainable material development, Lignin-based construction material, Recycling, X-ray computed tomography

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1. Introduction and Motivation

Sustainability has been an increasingly important focus for policymakers and industry leaders. The construction industry currently accounts for more than 11 % of global carbon emissions, the majority coming from the production of building materials such as concrete, which itself accounts for 8 % of annual man-made carbon emissions [1, 2]. To tackle the problem head on, new alternatives in the form of carbon-negative construction materials may be helpful. The development of construction materials that involve carbon sequestration is a powerful strategy for reducing the carbon footprint of the construction industry. Carbon sequestration takes place whenever bio-based construction materials, made from wood or agricultural residues, can be used, or when cement carbonation can be employed [3, 4, 5, 6]. The use of these novel materials helps limit the use of conventional construction materials produced through carbon dioxide emitting processes. Construction materials that sequester carbon also allow buildings to be used as carbon sinks which helps store carbon in the long-term [7]. Another avenue for further reduction of carbon emissions produced by the construction industry is through the development of methods for recycling these materials to achieve circular design/use. Circular design will help reduce demand for raw materials and limit waste generation [8]. Carbon sequestration and circularity are both beneficial.

The transition from linear to circular economies has been suggested as a model to promote sustainability through recycling (re-use of materials to produce new products) or up-cycling (re-use of materials to manufacture new superior products) at the end-of-life for a product [9, 10]. Circular design can be integrated into the production of construction materials, with linear processes (extracting raw resources → fabricating construction material → demolishing construction materials and sending to landfill) being replaced by circular processes (extracting raw resources → fabricating construction material → recycling construction material and avoiding landfill) [9]. Re-using construction materials at the end-of-life is an ideal method of preventing additional consumption of raw materials and reducing the generation of construction waste. An example of recycling in the construction industry is the re-use of recycled concrete aggregates [11].

Biopolymer-bound Soil Composites (BSC) are a new class of sustainable construction materials that utilize biopolymer binders to form composite materials. Biopolymers used in BSC include various proteins such as bovine serum albumin, animal blood plasma AP920, and polysaccharides,

such as xanthan gum [12, 13, 14, 15, 16, 17, 18, 19]. To fabricate BSC a solvent is used to dissolve the biopolymer, and the resulting solution is mixed with granular material to create a moldable mixture that can be formed into a variety of shapes. Solvent removal by desiccation results in a composite material that can be as strong as concrete [20].

Lignin is a hydrophobic biopolymer that can be used to make BSC. Its use in BSC will help provide increased durability against exposure to moisture or precipitation. Lignin is a strong candidate for reducing cement use, as it is the second most naturally occurring complex organic material globally, after cellulose. In countries with high levels of cement demand, lignin is already abundant, as it is a byproduct of existing pulping and bio-ethanol facilities [21, 22]. Local availability is important to consider when developing alternative bio-based materials, as it minimizes ecological and financial impacts associated with transportation. The use of carbon-rich lignin in making BSC leads to an additional benefit in the form of carbon sequestration, thus potentially making lignin-based BSC a carbon-negative construction material [23, 24, 6, 5, 25].

In this study, we describe lignin-based BSC made with two types of kraft lignin derived from the Lignoforce process (KL-1) and the Lignoboost process (KL-2) [26]⁴. Several mixes of lignin-based BSC were manufactured using the ingredients lignin (KL-1 and KL-2), sand, and solvent. We also manufactured lignin-based BSC using recycled lignin aggregate and solvent, and compared the results to lignin-based BSC that was manufactured using virgin ingredients. Additionally, we performed material characterization using X-ray micro-computed tomography, to analyze the microstructure of the two material variants. Our findings highlight the recyclability of lignin-based BSC and shed light on microstructural aspects of BSC that are enhanced by the recycling process.

2. Materials

To fabricate virgin lignin-based BSC, feldspathic grade 90 sand (# 90 Silver Sand; grain shape — sub-angular; hardness — 6.5 Mohs; moisture content — < 0.1 %; bulk density — 1409 kg/m^3 ; soil particle density — 2.60 g/cm^3 ; uniformity coefficient 1.75-1.95) from P.W. Gillibrand, Simi Valley, California was used. The maximum dry bulk density of this material ($\rho_{dry,bulksoil}^{max}$) was measured to

⁴Certain commercial equipment, software and/or materials are identified in this paper in order to adequately specify the experimental procedure. In no case does such identification imply recommendation or endorsement by the National Institute of Standards and Technology, nor does it imply that the equipment and/or materials used are necessarily the best available for the purpose.

be 1.48 g/cm^3 [20]. Lignoboost lignin (KL-1) was obtained from Stora Enso Oyj company (Helsinki, Finland), from their Sunila Mill. Lignoforce lignin (KL-2) was obtained from West Fraser Timber Company, Ltd (Vancouver, Canada), from their Hinton Plant, located in Alberta, Canada. The solvent used in this study was DMSO (reagent grade).

Previously published reports indicate that the molecular weight of softwood kraft lignin produced through the Lignoboost process is approximately 16,700 Da [27]. The lignoboost lignin described by Mattsson et al. is likely similar to the KL-1 lignin used in this study. The molecular weight of kraft lignin produced through the Lignoforce process is $5,426 \pm 29$ Da or $12,926 \pm 87$ Da for KL-2 lignin sourced from hardwood and softwood respectively [28]. Similarly, the lignoforce lignin described in Suota et al. is likely similar to the KL-2 lignin used in this study. In terms of elemental composition, KL-1 (from softwood, $C : 65.6\%$, $H : 5.7\%$, $N : 0.09\%$, $S : 1.85\%$, and $O : 26\%$) and KL-2 (from softwood, $C : 64.5\%$, $H : 5.7\%$, $N : 0.04\%$, $S : 1.70\%$, and $O : 28\%$ and from hardwood, $C : 62.9\%$, $H : 5.6\%$, $N : 0.06\%$, $S : 2.29\%$, and $O : 29\%$) are similar to one another when analyzed by a combustion elemental analyzer [28, 27, 29, 30].

3. Fabrication of Lignin-based BSC

Lignin was dissolved in DMSO by mixing in a stainless steel bowl with the aid of a stand mixer (household kitchen) for 5 min to 10 min. KL-1 lignin comes in a pelleted form and KL-2 lignin is supplied as a dry powder. Both types of lignin could be easily dissolved in DMSO. The resulting lignin-DMSO solution and grade 90 sand were mixed in the stand mixer for 5 minutes until fully blended. The wet mixture, similar in consistency to cookie dough, was shaped using a cylindrical molding apparatus (13 mm inner diameter and 76 mm height, composed of 316 stainless steel), to form cylindrical BSC samples. A steel piston (12.8 mm diameter, and 84 mm height) was inserted into the cylindrical mold to compact the samples. Double compaction (to 5 MPa) of the samples was performed using an MTS Criterion Model 43 electromechanical universal testing machine, which involved end-over-end compaction of each sample [20]. The compacted samples were extruded from the mold by pressing from one end using the piston. Immediately after extruding, the cylindrical samples were weighed and height measurements were obtained. Samples were then placed in a laboratory oven at 60°C to accomplish desiccation. The process of desiccation was tracked by measuring mass-loss over time. Upon complete desiccation, the samples were sanded on

both ends in order to ensure that the ends were parallel and at 90 degrees to the long axis of the specimens. Compressive strength testing was performed by uni-axial compression using the MTS universal testing machine described above. For more information regarding the virgin BSC mixes investigated in this study, please refer to Table 2 in the Supplementary Material.

Two types of kraft lignin — KL-1 and KL-2 — were used, with four mixes prepared for each type of lignin. Within each of these mixes, 7-8 cylindrical samples (height = $52.0 \pm 2.5 \text{ mm}$, & diameter = $25.4 \pm 0.5 \text{ mm}$) were fabricated. Measurements of dry bulk soil density, solvent content, and biopolymer content were made, and this data is summarized in Table 3 in the Supplementary Material.

The strength of BSC generally increases with higher content of biopolymer. For KL-2 lignin-based BSC the relationship between biopolymer content and compressive strength is not as strong as for KL-1 lignin-based BSC. Similarly, compressive strength of BSC increases with higher dry bulk soil density and dry density. Therefore, the optimal features for predicting lignin-based BSC strength are dry bulk soil density and biopolymer content. All of these relationships are illustrated in Figure 6 in the Supplementary Material.

Table 1: Models for Predicting the Compressive Strength of Lignin-Based BSCs

Fitted Models	Equations and Fit Parameters for Predicting BSC Strength		
Model-1	Equation 1	$\sigma_c = a * \%w_{BP/S} + c$ (1)	
	Fitted parameters	$\sigma_{KL-1} = [a = 33.7, c = 0.05]$ $\sigma_{KL-2} = [a = 32.9, c = -0.11]$	
Model-2	Equation 2	$\sigma_c = b * \rho_{dry,bulksoil} + c$ (2)	
	Fitted parameters	$\sigma_{KL-1} = [b = 44.3, c = -56.5]$ $\sigma_{KL-2} = [b = 31.1, d = -40.0]$	
Model-3	Equation 3	$\sigma_c = a * \rho_{dry,bulksoil} + b * \%w_{BP/S} + c$ (3)	
	Fitted parameters	$\sigma_{KL-1} = [a = 12.0, b = 23.3, c = -15.0]$ $\sigma_{KL-2} = [a = 25.4, b = 19.6, c = -34.5]$	
Types of Lignin BSCs	Performance of Fitted Models for Predicting BSC Strength		
	Model-1	Model-2	Model-3
KL-1	$R^2 = 0.99$	$R^2 = 0.89$	$R^2 = 0.96$
KL-2	$R^2 = 0.85$	$R^2 = 0.80$	$R^2 = 0.89$

The relationship between compressive strength and these key mix parameters can be accounted for by simple linear models. To fit the equations of Model-1, a zero point was added since BSC with

near zero biopolymer content will effectively have no biopolymer bridging and therefore have near zero compressive strength. We considered the possibility that a model including both dry bulk soil density and biopolymer content may be more effective in accounting for the compressive strength of lignin-based BSC. Model-3, therefore, was fitted for both types of kraft lignin. The fitted models and parameters are shown in Table 1, with Model-3 being the most effective in estimating the compressive strength of kraft lignin-based BSC. The resulting models and fit parameters are shown in Table 1.

The accuracy of these models were evaluated using the developed models to predict the compressive strengths of lignin-based BSC where the error in the predictions can be calculated by comparing the predicted strengths with the experimental results. Figure 7 (in the Supplementary Material) shows the predicted compressive strength results for KL-1 lignin-based BSC and KL-2 lignin-Based BSC in comparison with experimental results. Upon comparing the fitted models in Table 1 and Figure 7, the best model to compute the design area of the design guide is a bivariate model based on biopolymer content and dry bulk soil density (Model-3 in Equation 3).

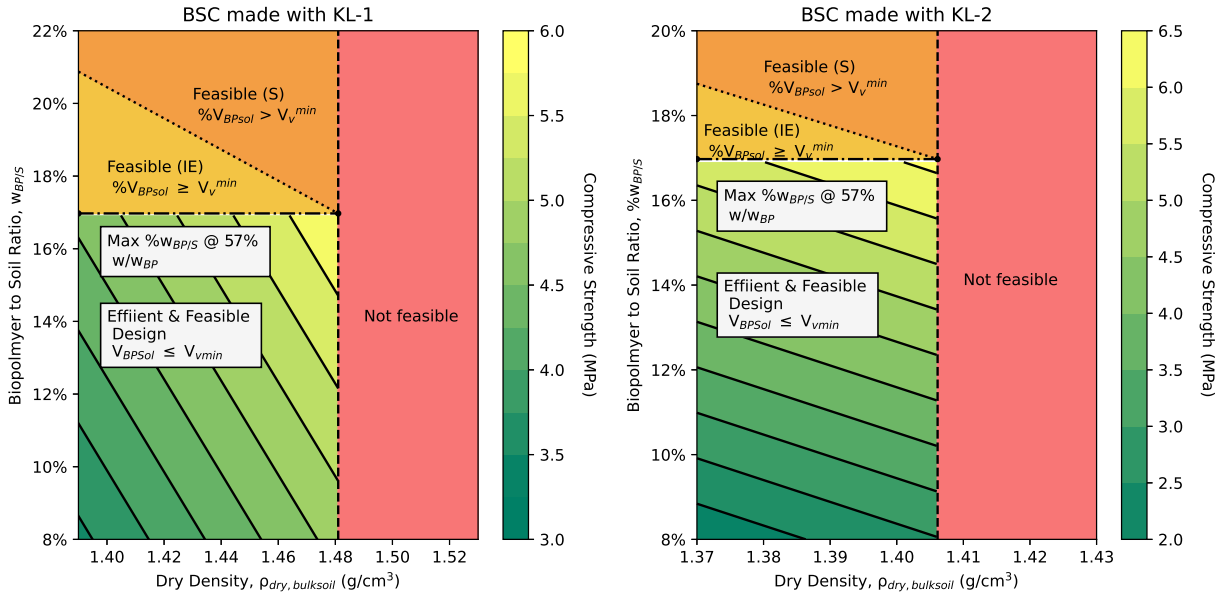


Figure 1: Design relationships for a) KL-1 lignin-based BSC and b) KL-2 lignin-based BSCs made from virgin ingredients. The contoured regions in the plots represent the efficient design area. The yellow and orange regions depict feasible design areas that are inefficient or more prone to shrinkage cracking respectively. The red region of the plot shows the infeasible design area, based on the fact that you cannot exceed the maximum dry bulk soil density.

Model-3 was used to plot the contoured design region in Figure 1, which represents the efficient and feasible designs for kraft lignin-based BSC. The contours shown in the figure represent the predicted compressive strength of lignin-based BSC for a set dry bulk soil density and biopolymer content. The yellow region represents designs that are feasible but inefficient (IE, volume of biopolymer solution exceeds the available void space). The orange region corresponds to designs that are feasible but are more prone to shrinkage cracking (SC). Finally, the red region represents the regions of the design relationships that cannot be fabricated, since no BSC samples can exceed the maximum dry bulk soil density.

4. Recycling of Lignin-Based BSC

The overall methodology (Figure 2) for the recycling of lignin-based BSC involves two main processes, (1) end-of-life handling of lignin-based BSC and (2) integration of recycled lignin aggregates to form recycled lignin-based BSC. Recycled lignin aggregates (RLA) are defined as particles of aggregate coated with lignin that are formed by mechanical disruption of lignin-based BSC products. In reality, the aggregate in RLA probably exists in the form of micro-clumps of aggregate together with lignin, rather than individual particles of aggregate. Sources of end-of-life (EOL) lignin-based BSC include BSC left over from fabrication, material harvested from finished structures that need modification, and damaged material at the end of design life. In the laboratory, RLA is produced by crushing previously manufactured lignin-based BSC with a sledgehammer for 10 minutes to 20 minutes, with the particulate matter being iteratively passed through sieves during the process. Sieving was used to ensure that RLA has consistent particle size for the recycling studies, with no residual matter being left on the 8 mm (5/16 in) sieve. The sieve process results are shown in Figure 8 (in the Supplementary Material). The results showed near identical sieve distributions for the various preparations of KL-1 RLA, but somewhat variable distributions for the various preparations of KL-2 RLA. For these preparations, all the RLA passed through the 0.3 inch (7.62 mm) sieve opening, thereby ensuring that RLA used to make recycled lignin-based BSC was reasonably consistent.

To produce recycled lignin-based BSC, a method similar to the procedure for fabricating virgin lignin-based BSC was implemented, with the primary difference being the substitution of virgin sand and lignin with RLA.

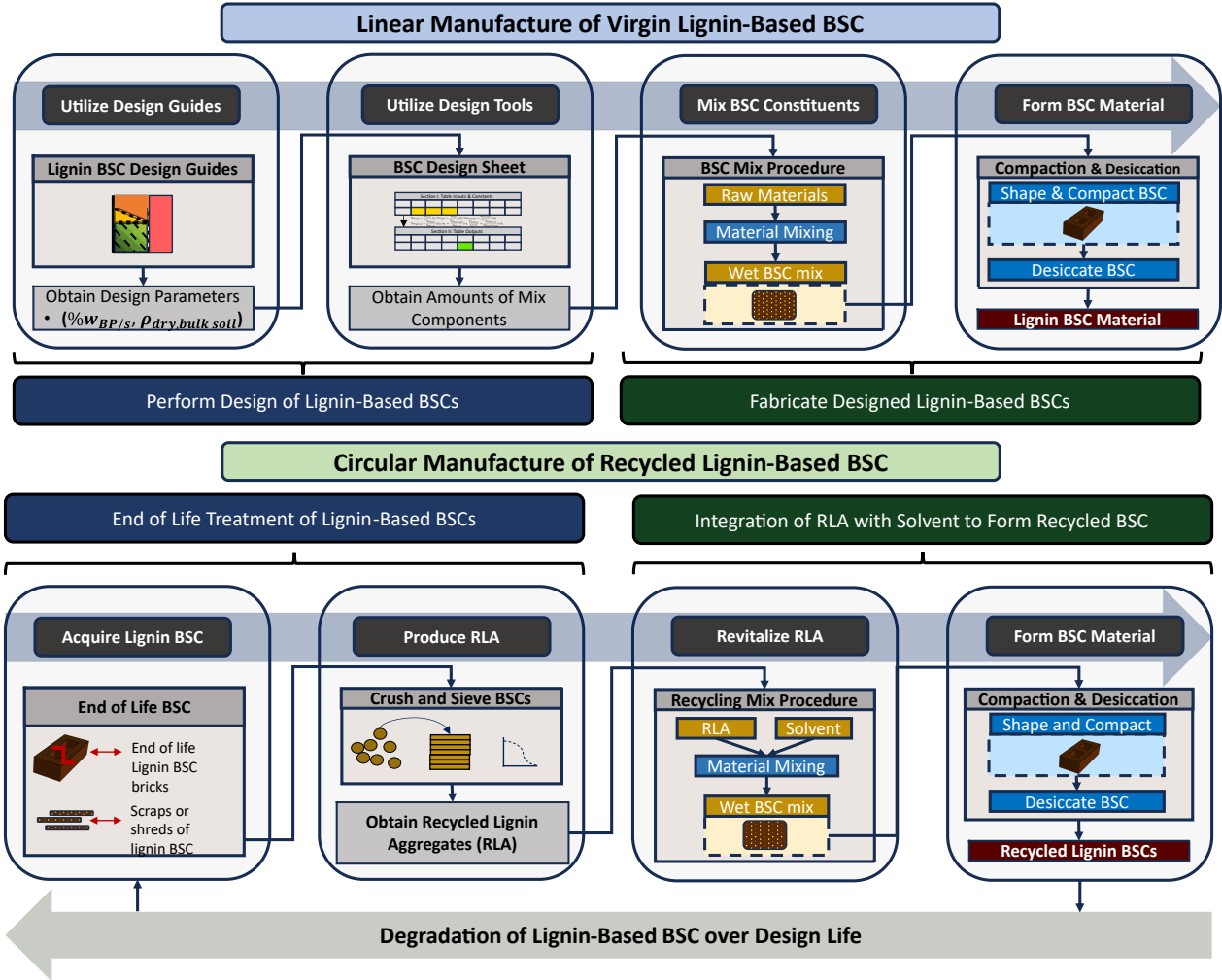


Figure 2: Scheme for linear and circular manufacture of lignin-based biopolymer-bound soil composites. Linear manufacture of lignin-based BSC uses virgin raw ingredients. Circular manufacturing involves: (1) crushing of lignin-based BSC (2) reconstituting the green mix using fresh solvent.

5. X-Ray Micro-CT Scan Analysis of Test Articles

X-ray micro-CT images (voxel size was $1.71\mu m$, also shown in Figure 4) of virgin lignin-based BSC and recycled lignin-based BSC (S90_DMSO_KL-1_BPS_50_BP_14, with diameter = 8 mm and height of 10 mm) were taken using a series of X-ray cone projection images produced using a Zeiss Versa XRM500, with the voltage and power set at 60 kV and 7 W, respectively. Even though the three phases of lignin-based BSC are distinguishable by visual inspection, the commercial software Dragonfly⁵ was used to segment the images into three material phases using convolutional neural

⁵The image segmentation for this paper was generated using Dragonfly software, Version 2022.2 for [Windows]. Comet Technologies Canada Inc., Montreal, Canada; software available at <https://www.theobjects.com/dragonfly>.

networks [31, 32]. For this work, we used a fully convolutional densely connected neural network (FC-DenseNet) approach for image segmentation [33, 34]. Further microstructure analysis was carried out by performing linear path function analysis and two-point probability analysis for each test article. Prior to computing the linear path and two-point probability functions for each dataset, the images were cropped to a size of 250 x 250 pixels.

From the cropped images we computed two geometric statistical descriptors: the linear path function and the two-point probability function [35]. The two-point probability function was used to analyze the relationship between the different phases (regions) of the material. Given that lignin-based BSC contains three phases (aggregate, void, and lignin biopolymer), there are three independent two-point probability correlation functions that can be calculated [36, 37]. We also implemented the linear path function, an additional geometric descriptor that is helpful for analyzing the biopolymer phase of the material. The linear path function describes the probability that a line of a given length (r in Figure 4) lies completely in one phase of the material [35, 20, 38, 36].

6. Results

Following the recycling scheme depicted in Figure 2, we created recycled lignin-based BSC test articles using mix #2 (made with KL-1 lignin). The compressive strength of the recycled product was 6.12 MPa compared to 4.38 MPa for its virgin counterpart (From Table 4 in the Supplementary Material). We examined another mix, mix # 6 (made with KL-2 lignin), and found that the compressive strength of recycled lignin-based BSC was 6.40 MPa which is significantly higher than the 2.87 MPa of its virgin counterpart (From Table 4 in the Supplementary Material). Since we did not observe additional strength gain with subsequent rounds of recycling, compressive strength data for all rounds of recycling were averaged.

To visualize these differences a series of histograms were prepared as shown in Figure 3. We compared the compressive strength of virgin KL-1 lignin-based BSC and virgin KL-2 lignin-based BSC, with their recycled counterparts that underwent up to four rounds of recycling.

To analyze the difference in compressive strength of lignin-based BSC made from virgin or recycled ingredients, a two-sample t-test was performed. The difference in compressive strength of virgin lignin-based BSC and recycled lignin-based BSC was significant at the 95% confidence level (p-value for KL-1 lignin-based BSC was 0.035 and p-value for KL-2 lignin-based BSC was $4.29e^{-6}$).

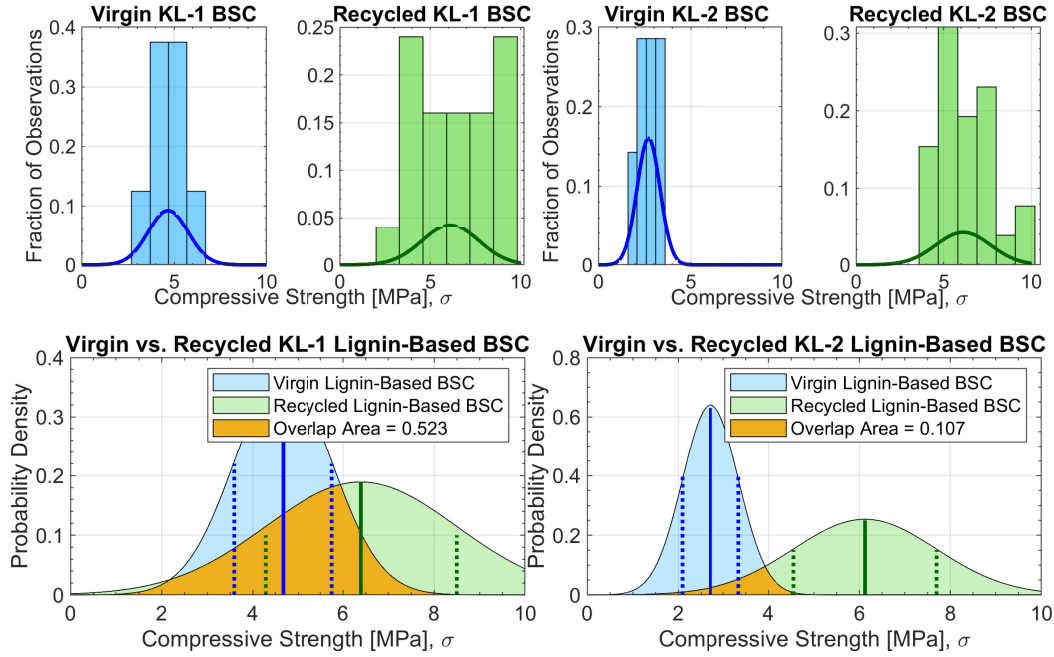


Figure 3: Recycled lignin-based BSC is stronger than lignin-based BSC made with virgin ingredients. The solid vertical lines in the lower panels represent the mean compressive strength, with the dotted lines representing the standard deviation.

6.1. Microstructure Analysis of Lignin-Based BSC

Given the significant difference in the compressive strength of recycled lignin-based BSC in comparison to the material made with virgin ingredients, we evaluated the microstructure of these biocomposites by performing X-ray micro-CT scan analysis on test articles made from the same mix designs. The micro-CT scan method allows us to distinguish between aggregate particles, biopolymer regions, and void spaces within the material. An efficient way to visualize the microstructure of the test articles is to create "slices", which can be computationally generated. Slices that are oriented parallel to the X-ray beam used for acquiring the data have the best resolution. The slices can be used for a variety of microstructure analyses.

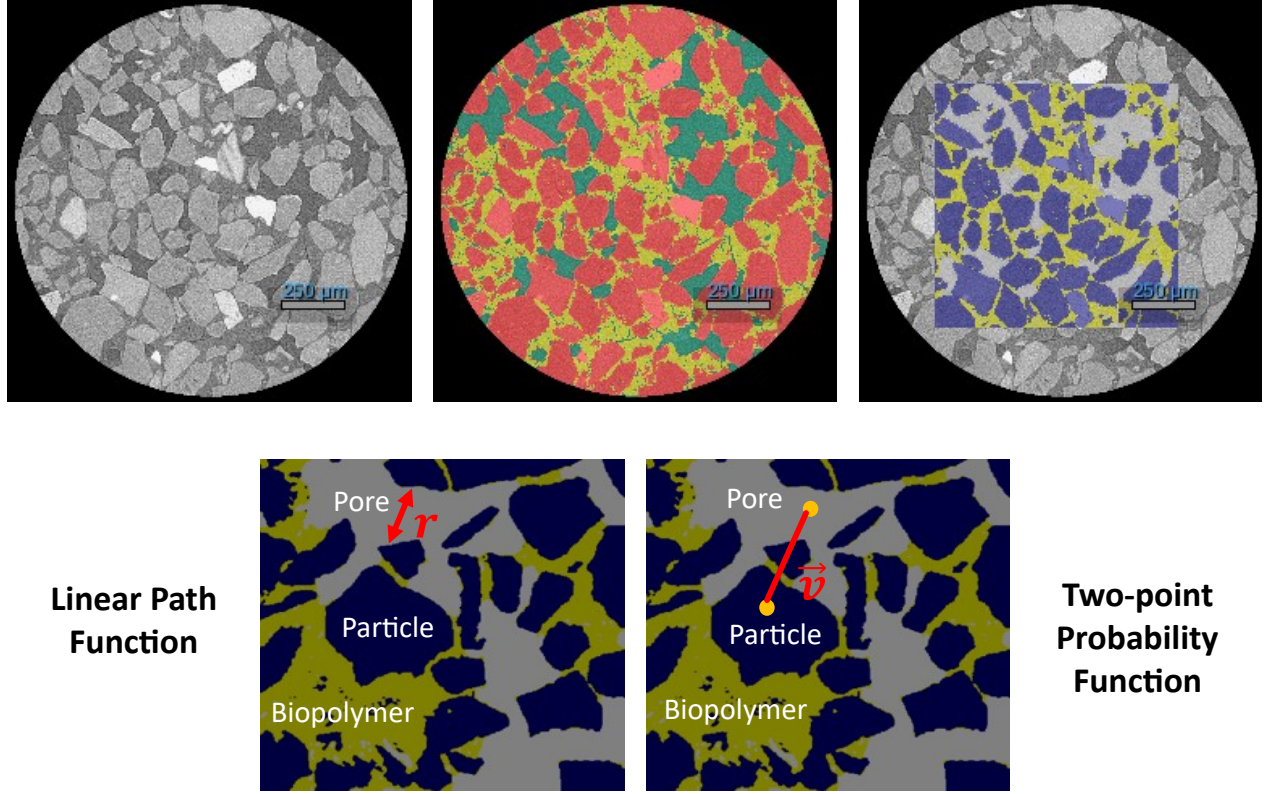


Figure 4: Overview of the microstructure analysis for lignin-based BSC using X-ray micro-CT. The images in the top row show the progression for image segmentation: raw micro-CT image $\rightarrow$ classified micro-CT image $\rightarrow$ cropped classified micro-CT image. In the classified micro-CT image, the aggregates are represented by the red areas; the teal areas depict the voids; and the yellow areas depict the lignin biopolymer. The zoomed-in images in the bottom row show the primary differences between the linear and two-point correlation functions.

For both the virgin lignin-based BSC sample and the recycled lignin-based BSC sample, a total of 993 2D slices were generated from the micro-CT data. Based on image segmentation for the virgin material, the volume fractions were: 57.6 % aggregate, 14.8 % biopolymer, and 27.6 % pores. For the recycled material, the volume fractions were: 56.1 % aggregate, 11.1 % biopolymer, and 32.7 % pores. Sample slices showing image segmentation can be seen in Figure 4. Because we only obtained a single micro-CT scan for each type of specimen (virgin lignin-based BSC and recycled lignin-based BSC), we were not able to perform a statistical analysis based on the volume fraction data. Instead we used two approaches for microstructure analysis, the linear path function and the two-point probability function, which together provide a comprehensive assessment of the microstructure of the specimens together with statistical parameters. The linear path function involves placing lines of arbitrary length and orientation into the micro-CT scan images, then counting the fraction of

those lines that lie wholly within a single phase of the material. The two-point probability function (correlation function) involves drawing lines between two points that can be identified within distinct phases of the material and correlating the distribution of lengths of these mathematically drawn lines.

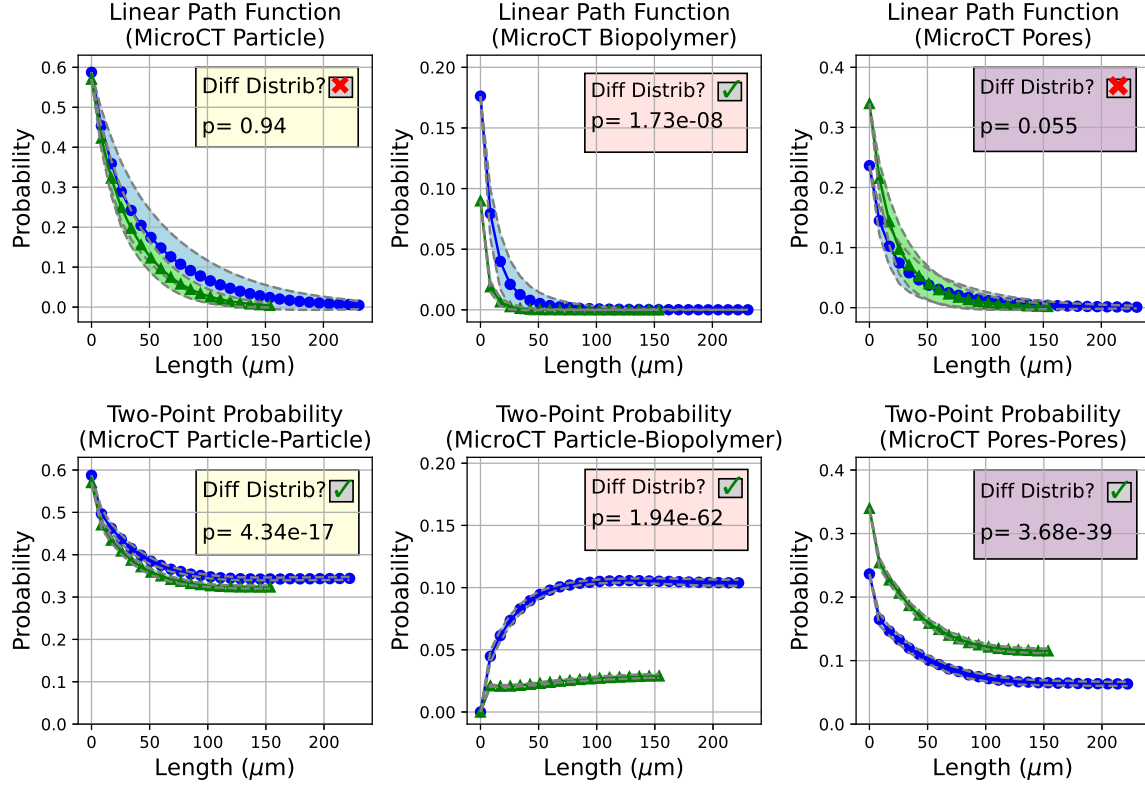


Figure 5: Comparison of the microstructure of virgin lignin-based BSC (blue circles/line) with the microstructure of recycled lignin-based BSC (green triangles/line), using the linear path function and two-point probability function. For each panel an envelope is drawn around the mean line, which shows the standard deviation for each descriptor. A Kolmogorov-Smirnov (K-S) test was computed for all three linear path functions and all three two-point probability functions. P-values are shown for each type of analysis.

The linear path function analysis revealed that the pores in the recycled lignin-based BSC were larger than the pores in the virgin material. This finding however did not reach statistical significance at the $p=0.05$ level, although it came close. The trend toward larger pores was consistent with an overall increase in pore fraction, as determined by image segmentation (virgin material: 27.6% pores and recycled material: 32.7% pores). Using the linear path function, we also observed

a shorter dimension for the biopolymer regions of the lignin-based BSC that was highly statistically significant. This finding indicates that the biopolymer is more tightly associated with the particles in the recycled lignin-based BSC compared to the virgin material.

For the two-point probability function, a striking reduction in the distance between an aggregate particle and adjacent biopolymer was seen. This observation is best explained by the concept that there was a higher degree of coating of each particle with biopolymer (lignin) in the recycled material. This makes sense as micro-clumps represent aggregate particles that are already partially covered with lignin, and they can acquire additional lignin coating as the material is mixed in the re-fabrication process.

7. Discussion of Results

Lignin-based BSC is a novel biocomposite that uses lignin as a binding agent to solidify granular material (aggregate) for construction purposes. The potential for recyclability of lignin-based BSC is based on the concept that the association between lignin and aggregate is reversible. Adding solvent to already-manufactured BSC products allows the lignin to be redissolved. By contrast, ordinary portland cement concrete is cured by an irreversible hydration reaction in the portland cement phase of the material. Thus, the portland cement binder in finished concrete cannot be recaptured for re-use. The prospect of recycling lignin-based BSC is attractive because it may further reduce the environmental impact of this material. This study presents an analysis of recycled lignin-based BSC in comparison to lignin-based BSC made from virgin ingredients.

Surprisingly, we discovered that lignin-based BSC made from recycled ingredients (recycled lignin aggregates, i.e. a material that is formed from mechanical disruption of lignin-based BSC, resulting in aggregate particles coated by lignin) had superior mechanical properties compared to lignin-based BSC made from virgin ingredients. This is unlike other materials, such as concrete, for which recycling typically leads to a weaker material [39, 40, 41]. Recycling of lignin-based BSC may decrease reliance on lignin and aggregate supply chains. This is especially important for construction at remote locations (terrestrially and extra-terrestrially), where transport is difficult and financially costly [42].

In this study, we examined the recyclability of lignin-based BSC looking at two different types of kraft lignin, KL-1 and KL-2. KL-1 is obtained from the Lignoboost process, while KL-2 is obtained

from the Lignoforce process. Although these two forms of kraft lignin are believed to be quite similar, a recently published study indicates that there are differences in molecular weight, with the KL-1 lignin having an estimated molecular weight of 16,700 Da and KL-2 lignin having an estimated molecular weight of 12,926 Da. In addition to differences in molecular weight, there may be differences in hydroxyl group content and differences in the branching structure of the lignin polymer. We have not attempted to perform our own characterization of these two types of lignin, but we have carefully evaluated the lignin-based BSCs made from these two ingredients, treating them as potentially different materials.

To understand the performance of these two types of kraft lignin-based BSC, we analyzed the relationship between compressive strength and biopolymer content of the biocomposite. The slopes in equation 1 for KL-1 lignin-based BSC and KL-2 lignin-based BSC were 33.7 and 32.9, respectively (see Figure 6 a) and Table 1). This small difference in the relationship between compressive strength and biopolymer content was not significant at the 95 % confidence level.

After a single round of recycling, lignin-based BSC made with KL-1 exhibited a 50% increase in compressive strength compared to virgin material. With additional rounds of recycling no further increase in compressive strength was seen. For KL-2, we observed an even greater enhancement of compressive strength with recycling. For lignin-based BSC made with KL-2, the first round of recycling resulted in a doubling of the compressive strength of the material. Again, as for BSC made with KL-1 lignin, with BSC made with KL-2 lignin, no further increase in compressive strength was seen with additional rounds of recycling.

We considered the possibility that the recycling process resulted in a change in the unit weight and/or dry bulk soil densities of the recycled lignin-based BSC test articles compared to the virgin lignin-based BSC test articles. The experimental results shown in Table 4 (in the Supplementary Material) indicate that the unit weight and dry bulk soil densities were roughly comparable for recycled lignin-based BSC after one round of recycling versus virgin lignin-based BSC. Furthermore, there was no trend in these parameters associated with subsequent rounds of recycling. When we compared virgin lignin-based BSC to recycled lignin-based BSC, we observed that the recycled material exhibited an increase in both unit weight and density. One possible explanation for the increase in density is that the recycled aggregate particles are already pre-coated with lignin, to some extent, prior to their use in the re-manufacturing process. More extensive coating of the

aggregate particles with lignin will likely enhance the formation of lignin bridges between particles. We favor this hypothesis as the explanation for the enhancement of the compressive strength of lignin-based BSC after recycling.

We also considered the possibility that the aggregate (sand), itself, might have been reduced in size because of the action of the sledgehammer used to break apart the virgin lignin-based BSC to start the recycling process. To investigate this possibility, we performed a sieve analysis of the RLA (shown in Figure 8 in the Supplementary Material) across all four rounds of recycling. The sieve analysis clearly indicates that the mechanical action of the sledgehammer resulted in the formation of micro-clumps of aggregate and lignin. We did not pulverize the virgin material sufficiently to regenerate individual sand grains.

We gained additional insight by performing a microstructure analysis by obtaining X-ray micro-CT scanned images of the recycled lignin-based BSC. We used two geometric descriptors — linear path function and two-point probability function — to characterize the microstructure comprehensively, using all the available image slices from the micro-CT data. These two methods are powerful ways of performing a microstructural characterization that uses all the available data in order to achieve an un-biased analysis.

The goal of the microstructure analysis was to understand the underlying reason for the enhancement in compressive strength that was observed after the test articles were recycled. The linear path function and the two-point probability function both indicate that the lignin biopolymer in the material becomes more closely associated with the aggregate particles during the process of re-manufacturing (recycling). We hypothesize that the mechanical disruption of lignin-based BSC generates micro-clumps of aggregate with associated biopolymer. When new solvent is added to remake the material, some of the lignin that is already associated with the aggregate particles remains in place, and additional lignin is able to further coat the aggregate particles. Thus, the re-manufacturing process increases the degree to which particles of aggregate become coated with lignin.

Its worthwhile to consider the possibility that lignin degradation may occur, depending on where this novel material is used, and this may affect recyclability [43]. Bio-degradation of lignin is well recognized, and this process can be enhanced in moist environments. In addition, acid rain and some components of air pollution potentially may lead to degradation of lignin-based BSC, especially on

the surface of manufactured products that are exposed to environment. While we can only speculate about degradation processes at this time, it may be helpful to point out that the optimum time for recapturing lignin from lignin-based BSC construction materials (at end of life) would be before any significant lignin degradation occurs.

Because lignin-based BSC uses a carbon-rich biopolymer, lignin-based BSC serves as a carbon sink [7]. Although lignin is considered a waste product, the most conservative allocation approach (in a life cycle assessment sense) considers that the carbon footprint associated with lignin production cannot be disregarded [25]. Thus, the recycling of lignin-based BSC, to recapture lignin, will serve to further minimize the carbon impact of lignin-based BSC. Furthermore, minimizing the demand for virgin lignin will reduce costs associated with the transportation of raw materials for the manufacture of lignin-based BSC construction materials. Since the recapture of lignin from lignin-based BSC is simple and straightforward, it is possible that small scale recycling operations could be established at medium to large scale construction sites to facilitate material re-use without having to rely on distant facilities. As lignin-based BSC construction materials are adopted for constructing the built environment, recycling methods should be integrated into the overall approach to using this material.

8. Conclusions

The development of BSC using a hydrophobic biopolymer such as lignin represents a significant step towards the use of bio-based materials as substitutes for conventional construction materials. This study introduces a new variant of lignin-based BSC, made with kraft lignin, that can be used for non-structural applications (e.g., pavers and non-load bearing walls). Additionally, we show that lignin-based BSC can be successfully recycled using a simple and easy to perform re-manufacturing process based on the fact that the association between biopolymer and aggregate is reversible. The transition from linear to circular design of lignin-based BSC is important as it may help to further reduce the carbon footprint of this new material that already looks attractive from an environmental impact perspective.

While the motivation for this study was directed at demonstrating recyclability of lignin-based BSC, we observed a surprising and unexpected improvement in the compressive strength of lignin-based BSC as a result of the re-manufacturing process. This observation of the improved compressive

strength serves to underscore the importance of biopolymer-aggregate association and its contribution to the mechanical properties of lignin-based BSC. These findings suggest that other variants of BSC that are manufactured using readily dissolved biopolymers may also be recyclable. It will be important to assess the mechanical properties of these other recycled BSC variants as well. Our findings suggest there may be opportunities to optimize the mechanical properties of lignin-based BSC and other forms of BSC, at the time of their initial manufacture, by taking steps to deliberately improve the association between the aggregate and the biopolymer, to create superior biocomposite materials.

This study makes the following contributions: (1) introduction of a method for recycling BSC that is relevant to lignin-based BSC and other types of BSC, (2) transitioning the design of BSC from conventional linear design into circular design, (3) providing a sustainable bio-based material that, with recyclability, may reduce the cost associated with transportation of raw ingredients, and (4) providing new insight into the microstructural features of BSC that contribute to its mechanical properties. Follow on studies examining the recyclability of other types of BSC are called for, which may help to further develop these novel biocomposites for application in the construction industry.

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Table 2: Experimental Mix Designs of Lignin-Based BSCs Using Virgin Lignin Biopolymer

Mix #	Mix Name*	%w _{soil}	%w _{BP}	%w _{DMSO}	%w _{BP/s}	Solv _c	%(w/w) _{BP}	$\frac{V_{BPsol}}{V_{min}}$
KL-1 Lignin								
1	S90_DMSO_KL-1_BPS_40_BP_11	78.0 %	8.8 %	13.2 %	11.3 %	16.9 %	40.0 %	0.83
2	S90_DMSO_KL-1_BPS_43_BP_12	77.9 %	9.6 %	12.5 %	12.3 %	16.0 %	43.5 %	0.83
3	S90_DMSO_KL-1_BPS_50_BP_14	77.8 %	11.1 %	11.1 %	14.3 %	14.3 %	50.0 %	0.83
4	S90_DMSO_KL-1_BPS_57_BP_16	77.4 %	12.7 %	9.8 %	16.5 %	12.7 %	56.5 %	0.83
KL-2 Lignin								
5	S90_DMSO_KL-2_BPS_40_BP_11	78.0 %	8.8 %	13.2 %	11.3 %	16.9 %	40.0 %	0.83
6	S90_DMSO_KL-2_BPS_43_BP_12	77.9 %	9.6 %	12.5 %	12.3 %	16.0 %	43.5 %	0.83
7	S90_DMSO_KL-2_BPS_50_BP_14	77.8 %	11.1 %	11.1 %	14.3 %	14.3 %	50.0 %	0.83
8	S90_DMSO_KL-2_BPS_57_BP_16	77.5 %	12.7 %	9.8 %	16.5 %	12.7 %	56.5 %	0.83

Table 3: Design and Compressive Strength of Lignin-Based BSC Mixes

Mix #	Mix Name	Fresh Unit Weight γ_{BSC}^{fresh} (g/cm ³)	Fresh Bulk Soil Density $\rho_{bulksoil}^{fresh}$ (g/cm ³)	Unit Weight γ_{BSC} (g/cm ³)	Dry Bulk Soil Density $\rho_{bulksoil}^{dry}$ (g/cm ³)	Compressive Strength σ_c (MPa)
KL-1 Lignin						
1	S90_DMSO_KL-1_BPS_40_BP_11	1.73 ± 0.04	1.36	1.52 ± 0.02	1.36	3.84 ± 0.43
2	S90_DMSO_KL-2_BPS_43_BP_12	1.74 ± 0.04	1.35	1.54 ± 0.03	1.37	4.38 ± 0.76
3	S90_DMSO_KL-1_BPS_50_BP_14	1.79 ± 0.04	1.39	1.59 ± 0.03	1.39	5.31 ± 1.31
4	S90_DMSO_KL-1_BPS_57_BP_16	1.77 ± 0.04	1.37	1.64 ± 0.02	1.41	5.73 ± 1.20
KL-2 Lignin						
5	S90_DMSO_KL-2_BPS_40_BP_11	1.76 ± 0.04	1.38	1.55 ± 0.01	1.39	3.54 ± 0.42
6	S90_DMSO_KL-2_BPS_43_BP_12	1.77 ± 0.04	1.38	1.58 ± 0.05	1.40	2.87 ± 0.52
7	S90_DMSO_KL-2_BPS_50_BP_14	1.91 ± 0.02	1.48	1.69 ± 0.02	1.48	5.95 ± 0.24
8	S90_DMSO_KL-2_BPS_57_BP_16	1.82 ± 0.01	1.41	1.66 ± 0.01	1.42	4.98 ± 0.66

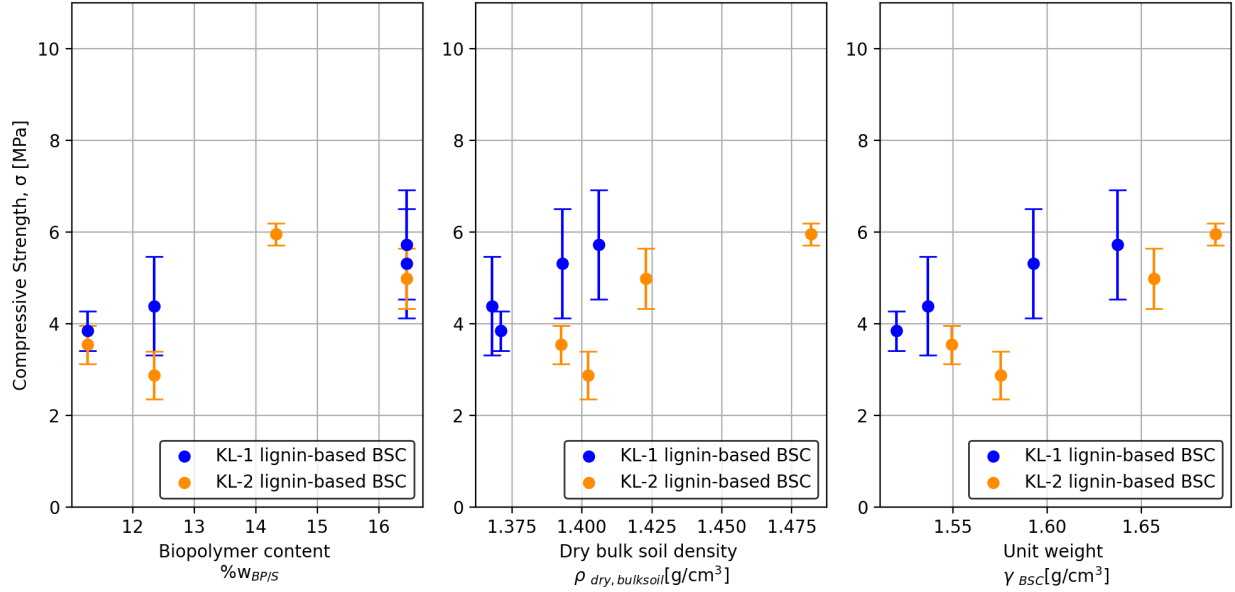


Figure 6: Compressive strengths of lignin-based BSC using KL-1 lignin and KL-2 lignin biopolymer as a function of (left $\rightarrow$ right) biopolymer content ($\%W_{BP/S}$), dry bulk soil density ($\rho_{dry, bulk soil}$), and unit weight. The error bars show the range of compressive strengths obtained at a given mix parameter. In this study around 7-8 samples were made at each mix parameter shown.

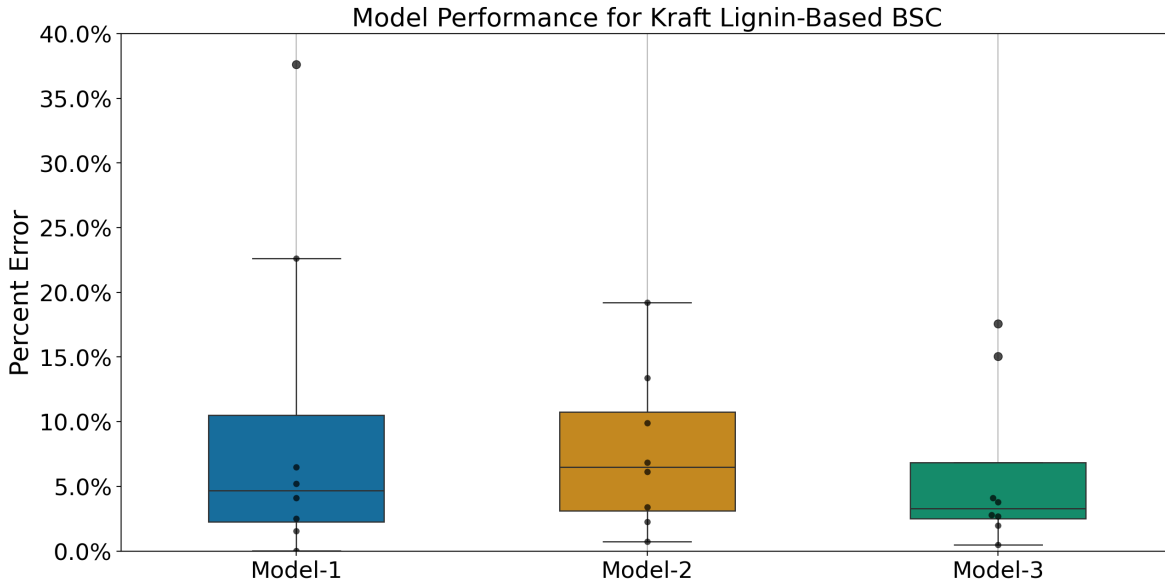


Figure 7: Overall percent error of compressive strength of kraft lignin-based BSC in comparison to predicted sample strengths for different regression models from equations 1-3 for two types of kraft lignin (KL-1 and KL-2).

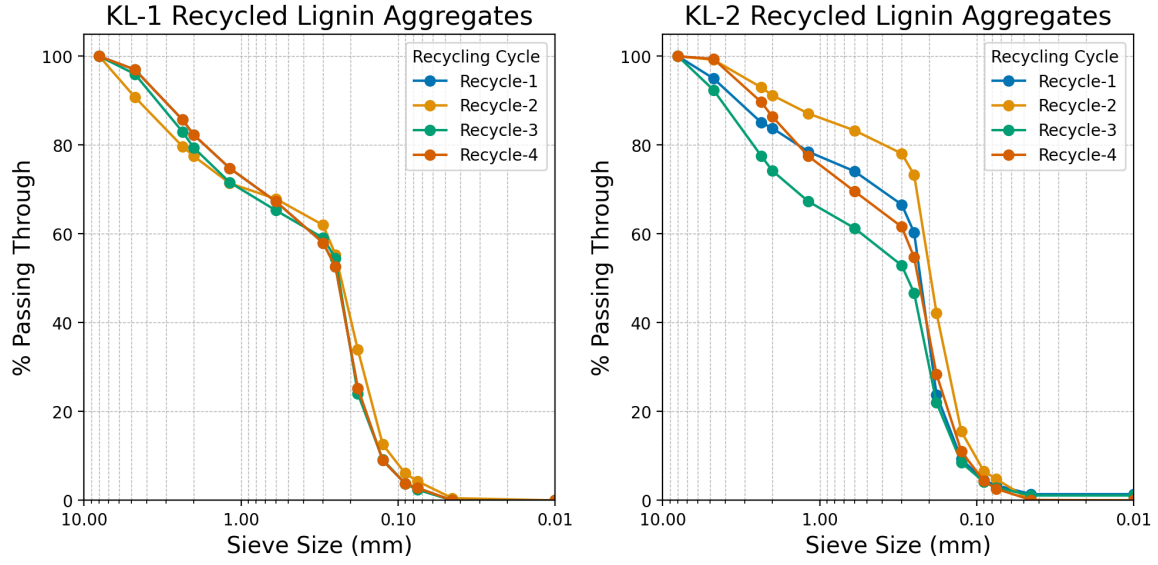


Figure 8: Sieve distributions of recycled lignin aggregates (RLA) used in making all recycled lignin-based BSC samples.

Table 4: Compressive strength results for Recycled Lignin-Based BSCs

Number of Times Recycled	Number of Samples n	Fresh Unit Weight γ_{BSC}^{fresh} (g/cm^3)	Fresh Bulk Soil Density $\rho_{bulksoil}^{fresh}$ (g/cm^3)	Unit Weight γ_{BSC} (g/cm^3)	Dry Bulk Soil Density $\rho_{bulksoil}^{dry}$ (g/cm^3)	Compressive Strength σ_c (MPa)
KL-1 Lignin made from [S90_DMSO_KL-1_BPS_43_BP_12]						
Virgin KL-1 BSC	7	1.74 ± 0.04	1.35	1.54 ± 0.03	1.37	4.38 ± 0.76
Recycle 1	7	1.82 ± 0.04	1.42	1.61 ± 0.03	1.44	6.39 ± 1.20
Recycle 2	7	1.81 ± 0.03	1.41	1.59 ± 0.02	1.41	4.90 ± 0.81
Recycle 3	6	1.84 ± 0.05	1.43	1.61 ± 0.01	1.44	7.57 ± 1.08
Recycle 4	6	1.80 ± 0.04	1.40	1.57 ± 0.01	1.40	5.00 ± 0.25
KL-2 Lignin made from [S90_DMSO_KL-2_BPS_43_BP_12]						
Virgin KL-2 BSC	7	1.77 ± 0.04	1.38	1.58 ± 0.05	1.40	2.87 ± 0.52
Recycle 1	7	1.83 ± 0.04	1.42	1.59 ± 0.02	1.42	7.54 ± 0.16
Recycle 2	7	1.81 ± 0.03	1.41	1.59 ± 0.03	1.42	5.74 ± 1.29
Recycle 3	6	1.82 ± 0.03	1.42	1.61 ± 0.01	1.43	3.87 ± 0.55
Recycle 4	5	1.88 ± 0.02	1.46	1.61 ± 0.02	1.44	8.74 ± 0.84

Note: The yellow highlights refer to mix # 2 and mix # 6 from Table 3

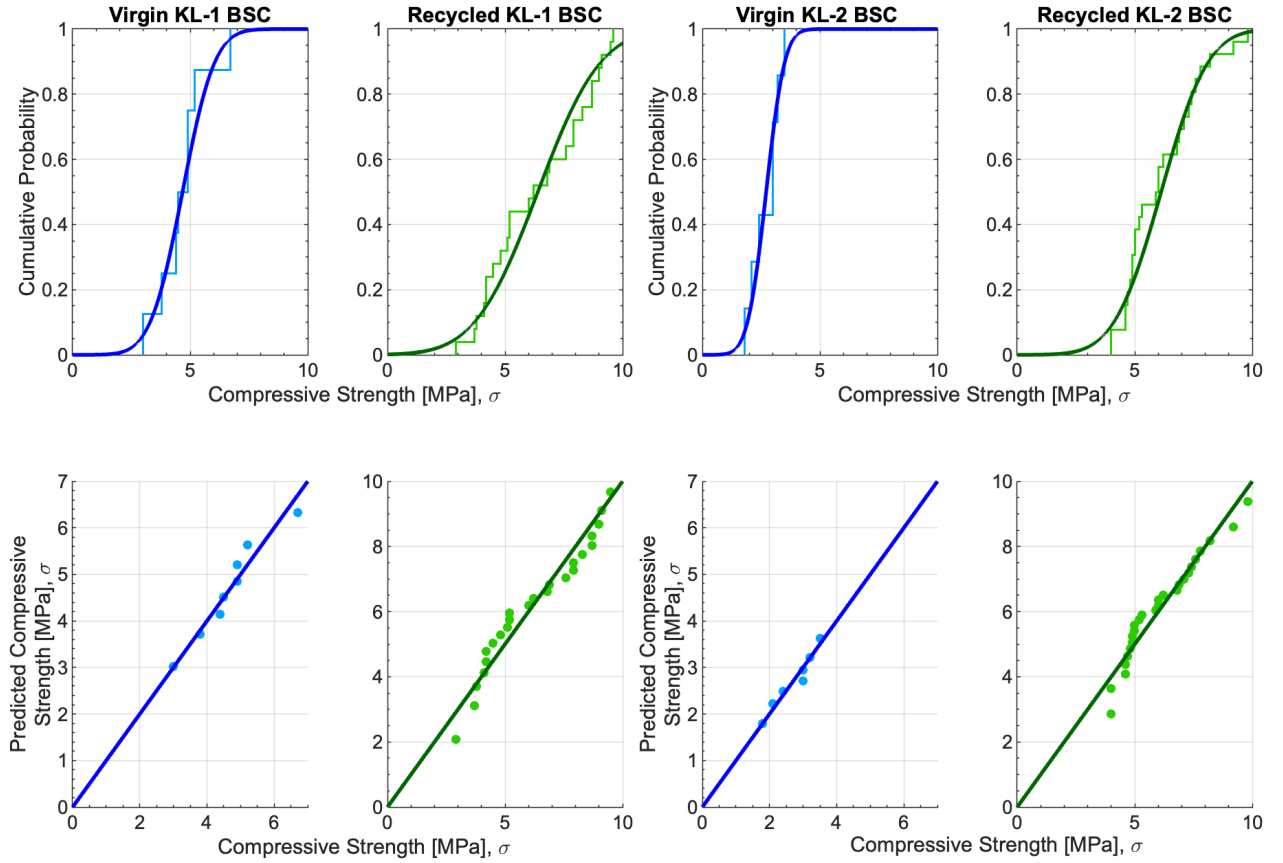


Figure 9: Statistical fits of compressive strength obtained for lignin-based BSC made with KL-1 and KL-2. The top row of figures shows the ECDF and CDF (with normal distribution) of lignin-based BSC made from virgin (shown in blue) and recycled (shown in green) ingredients. The bottom row of figures shows the QQ-plots of lignin-based BSC. This figure shows that the compressive strength results for virgin lignin-based BSC and recycled lignin-based BSC are normally distributed.